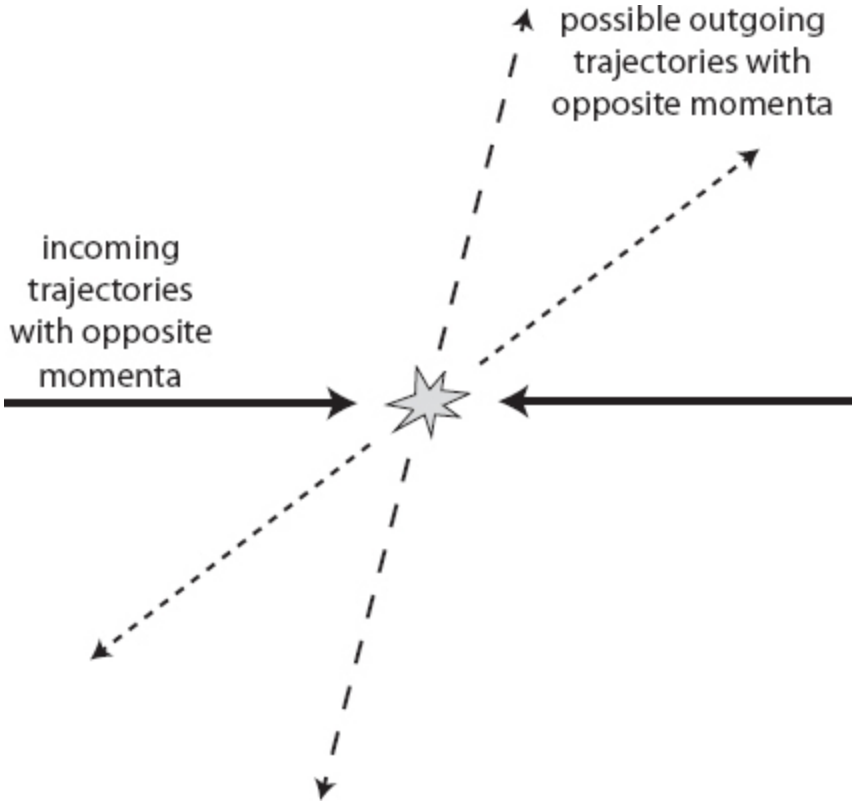
When we actually do this experiment, and observe the electrons after they have scattered, we notice something important. Since the electrons initially had equal and opposite velocities, the total momentum was zero. And momentum is conserved, so the post-interaction momentum should also be zero. This means that while the electrons might emerge moving in various different directions, whatever direction one of them moves in, the other moves in precisely the opposite.



That’s funny, when you think about it. The first electron has a probability of scattering at various angles, and so does the second one. But if they each had a separate wave function, those two probabilities would be completely unrelated. We could imagine just observing one of the electrons, and measuring the direction in which it’s moving. The other one would be undisturbed. How could it know that it’s supposed to be moving in the opposite direction when we actually do measure it?

We’ve already given away the answer. The two electrons don’t have separate wave functions; their behavior is described by the single wave function of the universe. In this case we can ignore the rest of the universe, and just focus in on these two electrons. But we can’t ignore one of the